

Homework 9

Marcel Gietzmann-Sanders

STAT651 - Statistical Theory I
University of Alaska Fairbanks

Question 1

a)

The function $g(x) = \sqrt{x}$ is concave and therefore:

$$g(\mathbb{E}(x)) \geq \mathbb{E}(g(x))$$

Therefore $g(\mathbb{E}(x))$ is an upper bound for $\mathbb{E}(g(x))$. As $\mathbb{E}(x) = 2.3(4.1) = 9.43$ our upper bound is $\sqrt{9.43} \approx 3.07$.

$$\begin{aligned}\mathbb{E}(\sqrt{x}) &= \int_0^\infty \sqrt{x} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} dx \\ &= \frac{\Gamma(\alpha + 0.5)\beta^{0.5}}{\Gamma(\alpha)} \int_0^\infty \frac{1}{\Gamma(\alpha + 0.5)\beta^{\alpha+0.5}} x^{\alpha+0.5-1} e^{-x/\beta} dx \\ &= \frac{\Gamma(\alpha + 0.5)\beta^{0.5}}{\Gamma(\alpha)} = \frac{\Gamma(2.8)\sqrt{4.1}}{\Gamma(2.3)} \approx 2.91\end{aligned}$$

b)

The function $g(x) = \ln x$ is concave and therefore:

$$g(\mathbb{E}(x)) \geq \mathbb{E}(g(x))$$

Therefore $g(\mathbb{E}(x))$ is an upper bound for $\mathbb{E}(g(x))$. As $\mathbb{E}(x) = 2.3(4.1) = 9.43$ our upper bound is $\ln 9.43 \approx 2.24$.

$$\mathbb{E}(\ln x) = \int_0^\infty \ln x \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} dx$$

According to wolfram alpha this is ≈ 2.01 .

Question 2

a)

First if $Y_i = aX_i$ then:

$$f_{Y_i}(y_i) = \frac{1}{\sqrt{2\pi}\sigma} e^{-y_i^2/(2a^2\sigma^2)} |1/a|$$

That is $Y_i \sim N(0, a^2\sigma^2)$.

So $Y_1 = -\frac{X_1}{\sqrt{2}\sigma} \sim N(0, 1/2)$ and $Y_2 = \frac{X_2}{\sqrt{2}\sigma} \sim N(0, 1/2)$. Let $W = Y_1 + Y_2$. Given these are independent we can look at the product of MGF's:

$$M_W(t) = M_{Y_1}(t)M_{Y_2}(t) = e^{t^2/4}e^{t^2/4} = e^{t^2/2}$$

Which means that $W \sim N(0, 1)$.

Now $U = W^2$ so that:

$$f_U(u) = \frac{1}{\sqrt{2\pi}} e^{-u/2} \frac{1}{\sqrt{u}}$$

Which is exactly the pdf of a chi-squared distribution with degrees of freedom = 1.

b)

The MGF for a $Z \sim \chi^2(k)$ distribution is $(1 - 2t)^{k/2}$. Therefore for a sum of independent $Z_i \sim \chi^2(k_i)$ distributions we have:

$$M_{\sum_i Z_i}(t) = \prod_i (1 - 2t)^{k_i/2} = (1 - 2t)^{\sum_i k_i/2}$$

That is:

$$W = \sum_i Z_i \sim \chi^2\left(\sum_i k_i\right)$$

In our specific case this means that a sum of m independent random variables that are each $\chi^2(1)$ distributed results in a random variable that is $\chi^2(m)$ distributed.

c)

Consider:

$$T/(2\sigma^2) = \sum_{i=1}^{n/2} \left(\frac{X_{2n} - X_{2n-1}}{\sqrt{2}\sigma} \right)^2$$

We already know that each of the terms in that summation are distributed as $\chi^2(1)$ from (a). Furthermore they are all independent as none of the X_j are shared between the terms. Therefore we have the sum of a set of independent random variables distributed as $\chi^2(1)$ which given (b) means:

$$T/(2\sigma^2) \sim \chi^2(n/2)$$

d)

The mean of a $\chi^2(m)$ distribution is just m . So in our case:

$$\mathbb{E}(T) = 2\sigma^2 n/2 = \sigma^2 n$$

e)

$$\mathbb{E}(W) = \mathbb{E}(T/n) = \mathbb{E}(T)/n = \sigma^2 n/n = \sigma^2$$

if $n > 0$. Therefore W is unbiased for σ^2 .

Question 3

a)

This is just a Beta($\theta + 1, 1$) distribution. Therefore $\mathbb{E}(X) = \frac{\theta+1}{\theta+2}$ and $\text{Var}(X) = \frac{\theta+1}{(\theta+2)^2(\theta+3)}$. From class we know that:

$$\mathbb{E}(\bar{X}) = \mathbb{E}(X) = \frac{\theta+1}{\theta+2}$$

$$\text{Var}(\bar{X}) = \text{Var}(X)/n = \frac{\theta+1}{n(\theta+2)^2(\theta+3)}$$

$\bar{X}$ is not unbiased for θ as $\mathbb{E}(\bar{X}) \neq \theta$.

b)

$$\mathbb{E}(\bar{X}) = \mathbb{E}(X) = \theta$$

$$\text{Var}(\bar{X}) = \text{Var}(X)/n = \theta/n$$

Here $\bar{X}$ is unbiased for θ as $\mathbb{E}(\bar{X}) = \theta$.

c)

Let's find k real quick.

$$\int_0^2 kx(x-2)dx = k \left(x^3/3 - x^2 \right) \Big|_0^2 = -4k/3 \rightarrow k = -3/4$$

$$\mathbb{E}(\bar{X}) = \mathbb{E}(X) = \int_0^2 xkx(x-2)dx = k \left(x^4/4 - 2x^3/3 \right) \Big|_0^2 = -4k/3 = 1$$

$$\mathbb{E}(X^2) = \int_0^2 x^2 kx(x-2)dx = k \left(x^5/5 - 2x^4/4 \right) \Big|_0^2 = -8k/5 = 6/5$$

$$\text{Var}(\bar{X}) = \text{Var}(X)/n = \frac{1}{n} (6/5 - (1)^2) = \frac{1}{5n}$$

Question 4

a)

From class we know that $\bar{X}_n \sim N(\mu_X, \sigma^2/n)$. The exact same steps demonstrate that $\bar{Y}_m \sim N(\mu_Y, \sigma^2/m)$.

b)

If $Z = \alpha U$ then we know that:

$$f_Z(z) = f_U(Z/\alpha)|1/\alpha|$$

Therefore: $-\bar{Y}_m \sim N(-\mu_Y, \sigma^2/m)$.

As the Y_i and X_i are independent $\bar{Y}_m$ and $\bar{X}_n$ are independent as well.
Therefore:

$$M_{\bar{X}_n - \bar{Y}_m}(t) = e^{\mu_X t + \sigma^2 t^2/n} e^{-\mu_Y t + \sigma^2 t^2/n} = e^{(\mu_X - \mu_Y)t + 2\sigma^2 t^2/n}$$

Therefore $\bar{X}_n - \bar{Y}_m \sim N(\mu_X - \mu_Y, 2\sigma^2/n)$

c)

From class we know that:

$$\frac{(n-1)S_X^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\frac{(m-1)S_Y^2}{\sigma^2} \sim \chi^2(m-1)$$

and from earlier in this homework we know the sum of two $\chi^2(q)$ independent random variables just results in a distribution that is also χ^2 but with the sum of degrees of freedom. That is:

$$\frac{(n-1)S_X^2}{\sigma^2} + \frac{(m-1)S_Y^2}{\sigma^2} \sim \chi^2(m+n-2)$$

Question 5

Let $Z = T^2$ then:

$$f_Z(z) = f_T(\sqrt{z}) \frac{1}{2\sqrt{z}} = k \left(1 + \frac{1}{p}z\right)^{-(p+1)/2} \frac{1}{2\sqrt{z}} = \frac{k}{2} \frac{z^{1/2-1}}{(1 + (1/p)z)^{(1+p)/2}}$$

Which is the same functional form as $F(1, p)$. Therefore knowing that $k/2$ will result in the normalization of this function into a pdf $T^2 \sim F(1, p)$.

Question 6

By definition we know that if $X \sim F(p, q)$ then there exists a $U \sim \chi^2(p)$ and $V \sim \chi^2(q)$ s.t.

$$X = \frac{U/p}{V/q}$$

so that $X \sim F(p, q)$.

In that case: $Y = 1/X = \frac{V/q}{U/p}$. Therefore it follows that $Y \sim F(q, p)$.